

Reactive Oxygen Species Detection with Fluorescent Probes: Limitations and Recommendations beyond DCFH-DA

Johannes Karges*

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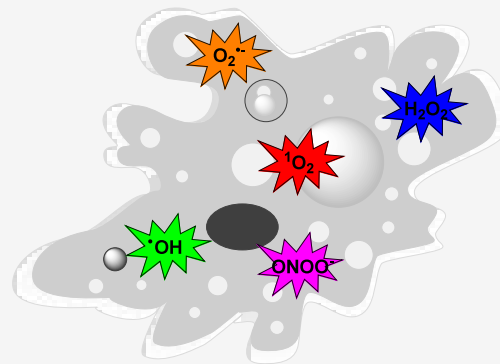
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ABSTRACT: Reactive oxygen species (ROS) are highly reactive molecules derived from molecular oxygen that play critical roles in cellular signaling, homeostasis, and the regulation of various physiological and pathological processes. Accurate detection and quantification of ROS are essential for elucidating their roles in health and disease. Despite significant advances in ROS probe development, challenges persist due to their short lifetimes, high reactivity, and their chemical diversity. This perspective article provides a critical evaluation of the limitations of the most commonly applied probe molecule DCFH-DA. Capitalizing on this, recommendations with practical applications in the lab for the specific detection of hydrogen peroxide, hydroxyl radical, singlet oxygen, superoxide anion, and peroxynitrite are provided. By integrating current knowledge on ROS probe technologies, this work aims to guide researchers in reliably assessing ROS in complex biological systems, thereby facilitating a deeper understanding of ROS-mediated processes and their implications for disease research and therapeutic development.



INTRODUCTION

Reactive oxygen species (ROS) are highly reactive molecules derived from molecular oxygen that play diverse roles in cellular physiology and pathology.^{1–3} Key members of this group include hydrogen peroxide (H_2O_2), hydroxyl radical ($\cdot\text{OH}$), singlet oxygen ($^1\text{O}_2$), superoxide anion ($\text{O}_2^{\cdot-}$), and peroxynitrite ($\text{ONOO}^-/\text{ONOOH}$), which are commonly studied due to their biological relevance and distinct chemical reactivity (Table 1).^{4–8} While historically considered primarily as damaging byproducts of cellular metabolism, it is now well-established that ROS play complex and context-dependent roles in both normal physiology and disease. At low to moderate levels, ROS are integral to cell signaling pathways that regulate proliferation, differentiation, immune responses, and cell death. However, excessive or uncontrolled ROS production, often referred to as oxidative stress, can damage cellular components such as DNA, proteins, and lipids, contributing to the pathogenesis of various diseases including cancer, neurodegenerative disorders, cardiovascular diseases, and inflammatory conditions.^{9–13}

In the context of drug discovery, ROS have emerged as both therapeutic targets and mechanistic effectors. Numerous therapeutic strategies now aim to either modulate ROS levels or exploit oxidative stress as a means of selectively targeting diseased cells. For example, many chemotherapeutic agents, including metal-based complexes, kinase inhibitors, and photodynamic therapy agents, induce cytotoxicity through ROS generation, leveraging the heightened oxidative stress sensitivity of cancer cells. Similarly, antioxidants are being developed to counteract ROS-associated tissue damage in diseases such as

Alzheimer's, Parkinson's, and atherosclerosis.^{14–21} Increasing number of studies suggest that ROS are crucial for physiological signaling regulating migration,^{22,23} differentiation,^{24,25} and the immune system.^{26–28}

Despite their importance, ROS are inherently challenging to detect due to their short half-lives, low concentrations, and high reactivity. Moreover, the cellular microenvironment, including pH, metal ion availability, and the presence of antioxidants, can influence ROS dynamics and confound measurements. To address these challenges, a variety of analytical methods have been developed, including electron spin resonance, chemiluminescence, colorimetric assays, and most commonly, fluorescent probes. However, several fluorescent probes come with significant limitations. These include nonspecificity for particular ROS species, susceptibility to auto-oxidation, probe instability, and dependence on intracellular enzymatic conversion, which can vary between cell types and experimental conditions. Misinterpretation of probe-derived fluorescence can lead to inaccurate conclusions regarding the activity of a compound, potentially misleading the assessments of the mechanism of action or therapeutic effect of a compound.

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Table 1. Overview of Biologically Most Relevant Types of ROS

type of ROS	chemical formula	reactivity
hydrogen peroxide	H ₂ O ₂	limited interaction with most biological molecules: hydrogen peroxide generally does not react with the majority of biomolecules generation of hydroxyl radicals: when exposed to specific transition metals, hydrogen peroxide can rapidly produce highly reactive hydroxyl radicals key biological targets: its primary interactions in biological systems involve heme groups, thiols, and peroxidase enzymes formation of a more reactive derivative: it combines with carbon dioxide to generate peroxymonocarbonate a compound with greater reactivity
hydroxyl radical	•OH	limited selectivity: damages whatever biological structures it encounters first rapid reactions: it attacks nearby molecules almost instantly at near diffusion-controlled rates
singlet oxygen	¹ O ₂	limited lifetime and diffusion range: singlet oxygen is a short-lived ROS with a limited diffusion radius, restricting its action to nearby biomolecules key biological targets: its primary targets include membrane lipids, specific amino acid residues in proteins, and nucleobases in DNA and RNA formation by energy transfer: singlet oxygen is typically generated by energy transfer from a photosensitizer to molecular oxygen
superoxide anion	O ₂ • ⁻	limited reactivity with biomolecules: the superoxide radical anion is relatively selective and does not degrade most cellular components reduction of metal ions: it can reduce transition metals like iron Fe³⁺ or copper Cu²⁺ ultrafast reaction with nitric oxide: it combines almost instantaneously with nitric oxide, forming peroxynitrite generation of hydroperoxides: it reacts with other free radicals in the presence of protons to produce organic hydroperoxides, which can propagate oxidative stress disruption of iron–sulfur enzymes: it specifically targets and damages enzymes with Fe–S clusters, impairing critical metabolic and redox functions
peroxynitrite	ONOO ⁻ /ONOOH	rapid interaction with thiols and metals: peroxynitrite quickly reacts with sulfur-containing compounds and transition metal centers dual damaging pathways: acts both as a direct oxidant and as a source of secondary free radicals radical formation by breakdown: decomposition of peroxynitrous acid produces highly reactive hydroxyl radicals carbon dioxide dependent transformation: it combines with carbon dioxide to form nitrosoperoxycarbonate (ONOCO₂⁻), altering its reactivity

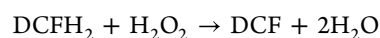
Consequently, researchers must use these tools with a clear understanding of their chemical properties, detection mechanisms, and experimental constraints. Capitalizing on this, this perspective article offers a critical assessment of the most commonly used fluorescent probes for detecting ROS along with practical recommendations for selecting appropriate probes for specific ROS types. Rather than providing a comprehensive literature review, which are available in prior reviews,^{29–34} this article is intended as a practical guide for nonspecialists to the field. It highlights the advantages and limitations of each probe, outlines practical considerations for their effective selection of the right ROS probe for its intended application.

■ DCFH₂-DA AS ROS PROBE

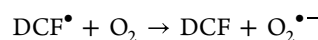
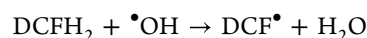
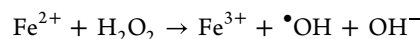
DCFH₂-DA is a widely used probe for detecting intracellular oxidative stress due to its cell permeability and fluorescent readout. After diffusing into cells, DCFH₂-DA is hydrolyzed by intracellular esterases to yield DCFH₂, a nonfluorescent and membrane-impermeable compound that accumulates in the cytosol. In the presence of reactive oxygen and nitrogen species, DCFH₂ is oxidized to the highly fluorescent dichlorofluorescein (DCF). However, this oxidation is not specific to a single ROS. Instead, it reflects the cumulative activity of several reactive species (Figure 1). Therefore, DCF fluorescence serves as a general indicator of oxidative activity rather than a selective measure of any individual ROS type.^{35–38} Below an examination of the detection of different types of ROS is listed:

- **Detection of H₂O₂:** Despite its central role in redox biology, H₂O₂ does not directly oxidize DCFH₂ to the fluorescent DCF product. In aqueous systems devoid of catalysts, H₂O₂ exhibits low reactivity toward DCFH₂. However, in the presence of peroxidase enzymes, such as horseradish peroxidase, cytochrome c, or myeloperoxidase, or transition metals like Fe(II) or Cu(II), H₂O₂ serves as a substrate to generate highly reactive intermediates or hydroxyl radicals through Fenton-type chemistry. These intermediates are capable of oxidizing DCFH₂. As such, the oxidation of DCFH₂ in response to H₂O₂ is highly context-dependent, relying heavily on the presence and activity of catalysts. Therefore, DCF fluorescence should not be equated directly with H₂O₂ levels unless the catalytic conditions are precisely defined and controlled.^{39–42}

Enzymatic mediated:



Fenton Reaction mediated:



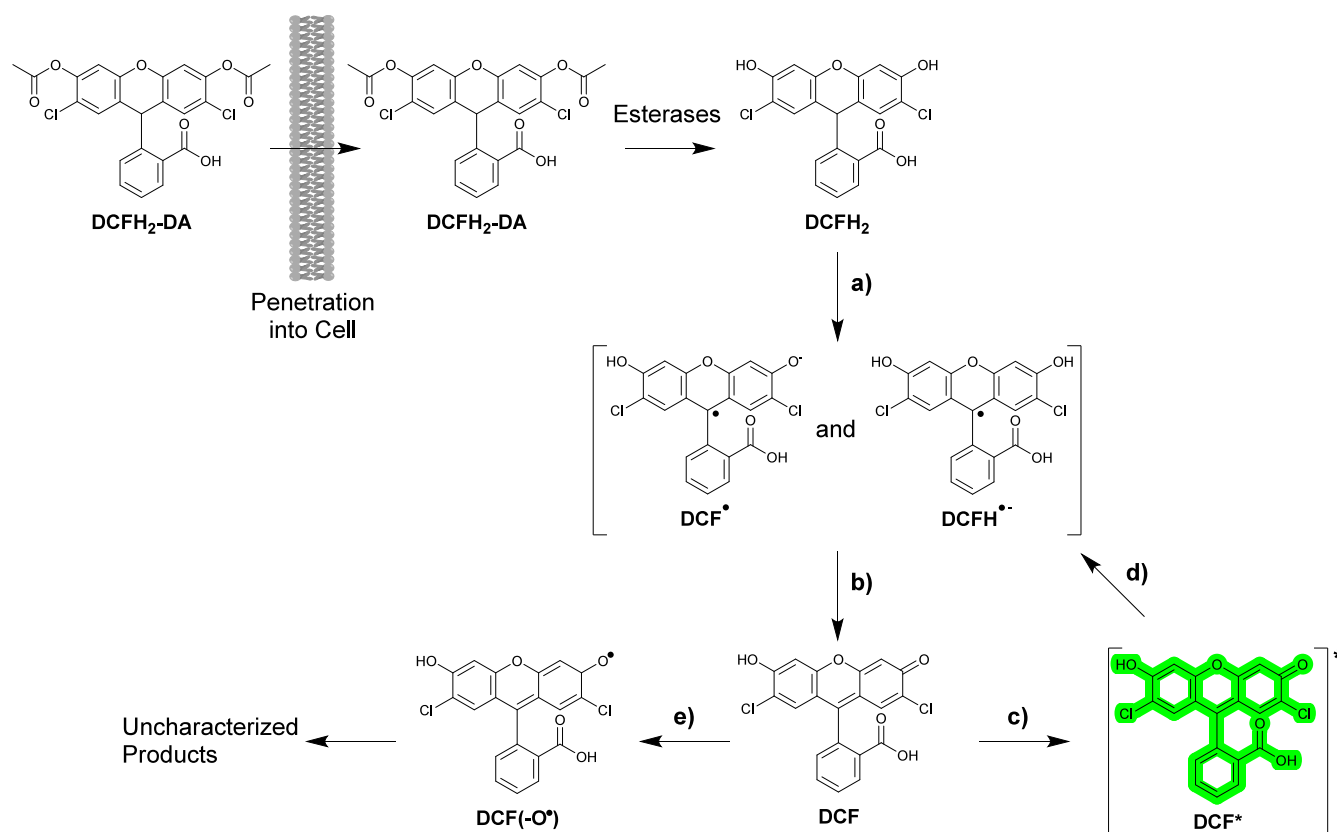
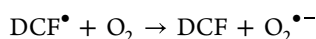
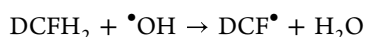


Figure 1. Mechanistic overview of the cellular oxidation pathway of DCFH₂-DA from passive membrane diffusion and intracellular esterase-mediated deacetylation to oxidation by a spectrum of reactive oxygen and nitrogen species, leading to the formation of fluorescent DCF, a widely employed but nonspecific indicator of oxidative stress. Conditions: (a) $\cdot\text{OH}$; $\text{H}_2\text{O}_2/\text{Fe}^{2+}$; $\text{O}_2\cdot^-$; $\text{ONOO}^-/\text{ONOOH}$; heme proteins; glutathione radicals, (b) O_2 , (c) light, (d) glutathione; nicotinamide adenine dinucleotide, (e) $\cdot\text{NO}_2$ or $\text{CO}_3\cdot^-$.

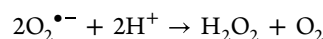
- **Detection of $\cdot\text{OH}$:** Among the most reactive ROS, the hydroxyl radical readily and rapidly oxidizes DCFH₂ in a diffusion-limited manner. Due to its extreme reactivity and lack of selectivity, $\cdot\text{OH}$ can interact with a wide array of cellular constituents, including aromatic rings and nucleic acids. In the context of DCFH₂, $\cdot\text{OH}$ efficiently converts the nonfluorescent molecule into its fluorescent derivative. However, the resulting oxidation products may include side products or partially oxidized forms, depending on the reaction site and conditions. Importantly, because $\cdot\text{OH}$ is so reactive, its biological half-life is extremely short, and its reactivity is largely dictated by proximity. Thus, unless DCFH₂ is present in high concentrations or $\cdot\text{OH}$ is generated in immediate vicinity to the probe, the extent of oxidation may be limited in cellular systems. Nonetheless, when $\cdot\text{OH}$ is the dominant species, DCFH₂ serves as a relatively effective reporter of oxidative events.^{43–45}



- **Detection of $^1\text{O}_2$:** Singlet oxygen, a form of molecular oxygen in an electronically excited state, participates in a range of oxidative signaling and damage pathways,

particularly under photooxidative stress. The interaction of $^1\text{O}_2$ with DCFH₂ is less well characterized and appears to be inefficient under physiological conditions. Some oxidation of DCFH₂ will occur in systems where $^1\text{O}_2$ is generated, but the reaction is neither rapid nor specific. Compared to other oxidants such as $\cdot\text{OH}$ or ONOO^- decomposition products, $^1\text{O}_2$ contributes minimally to DCF fluorescence. Consequently, DCFH₂-DA is able to detect $^1\text{O}_2$ but ideally should not be used as a primary probe for singlet oxygen detection.^{46–48}

- **Detection of $\text{O}_2\cdot^-$:** The superoxide anion, a primary ROS produced by various enzymatic and nonenzymatic reactions, shows very limited direct interaction with DCFH₂. Experimental studies have demonstrated that even when superoxide is generated in a controlled, quantifiable manner, such as by xanthine oxidase systems or by radiolytic methods, it produces minimal oxidation of DCFH₂. Moreover, the inclusion of superoxide dismutase (SOD), which catalyzes the dismutation of $\text{O}_2\cdot^-$ to H_2O_2 , typically has no effect on DCF fluorescence, reinforcing the notion that $\text{O}_2\cdot^-$ does not directly contribute to DCFH₂ oxidation. Any signal observed in superoxide-generating systems likely arises from secondary reactions, such as the conversion of superoxide to H_2O_2 followed by metal- or peroxidase-catalyzed oxidation of the probe.^{44,45,49}



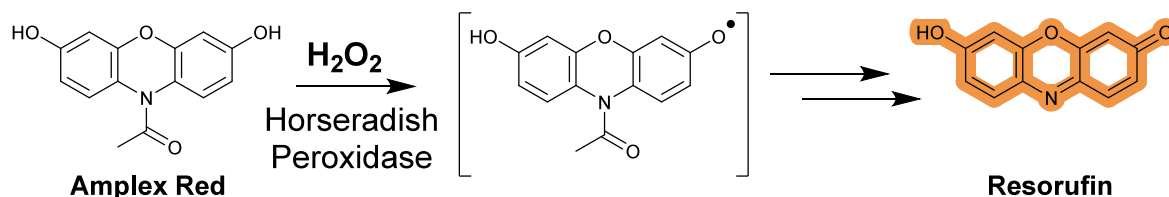
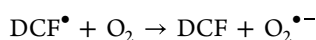
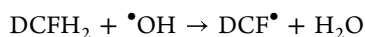
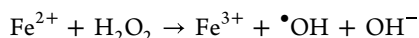
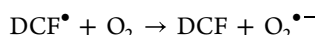
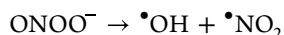


Figure 2. Simplified mechanism of action of Amplex Red toward the detection of H_2O_2 in the presence of horseradish peroxidase.



- **Detection of ONOO^- :** Peroxynitrite is a potent oxidant and nitrating agent formed by the diffusion-controlled reaction of superoxide with nitric oxide. While it was initially proposed that ONOO^- directly oxidizes DCFH_2 , subsequent kinetic analyses suggest that the observed oxidation is largely mediated by secondary radicals generated during peroxynitrite decomposition. These include hydroxyl radicals, nitrogen dioxide ($\bullet\text{NO}_2$), and carbonate radicals ($\text{CO}_3^{\bullet-}$), particularly in the presence of physiological concentrations of bicarbonate. Therefore, in systems where peroxynitrite is involved, DCFH_2 oxidation is best interpreted as a readout of radical decomposition products, not the parent molecule itself.^{50–53}



■ LIMITATIONS OF DCFH_2 -DA

Although DCFH_2 -DA is frequently employed to investigate ROS-related cellular events, its reliability as a selective indicator is flawed. The redox chemistry underlying its fluorescence activation is intricate and can lead to multiple sources of error and misinterpretation of data. Researchers must therefore exercise caution when using this probe and carefully consider the associated chemical caveats:

- **Lack of Direct Reactivity with H_2O_2 :** Contrary to common assumptions, DCFH_2 is not directly oxidized by H_2O_2 . Instead, the oxidation process typically requires the presence of catalytic intermediates such as peroxidases or redox-active metal ions. As a result, DCF fluorescence should not be interpreted as a direct measure of intracellular H_2O_2 concentrations.

- **Nonspecific Oxidation by Diverse Reactive Species:** DCFH_2 is highly susceptible to oxidation by a variety of one-electron oxidants beyond H_2O_2 . These include $\bullet\text{OH}$; $\text{H}_2\text{O}_2/\text{Fe}^{2+}$; $\text{O}_2^{\bullet-}$; $\text{ONOO}^-/\text{ONOOH}$; heme proteins; glutathione radicals.
- **Redox Cycling and Signal Amplification:** The intermediately produced radical species $\text{DCF}^\bullet$ generated during the oxidation of DCFH_2 can react rapidly with molecular oxygen, forming $\text{O}_2^{\bullet-}$. Superoxide dismutation produces additional H_2O_2 , potentially initiating a self-amplifying redox cycle. This cycle may artificially inflate fluorescence readings and obscure the true extent and nature of ROS generation under experimental conditions.
- **Influence of Apoptosis and Cytochrome c:** During apoptotic events, cytochrome c is released into the cytosol and may act as a peroxidase-like catalyst, facilitating DCFH_2 oxidation. Consequently, increased DCF fluorescence in apoptotic cells may be erroneously attributed to elevated ROS levels, when in fact it may reflect changes in peroxidase activity or mitochondrial disruption.
- **Contribution of Redox-Active Metal Ions:** Transition metals such as Fe^{2+} can accelerate DCFH_2 oxidation through Fenton-type reactions in the presence of oxygen or H_2O_2 . These metal-mediated pathways can confound attempts to quantify ROS, especially in systems with dysregulated iron homeostasis or exposure to metal-based therapeutics.
- **Variability Between Experimental Conditions:** Differences in probe uptake, esterase activity, and intracellular conditions between control and treated cells can significantly influence the rate of DCF generation and the efficiency of redox cycling. Thus, even when changes in fluorescence are observed between experimental groups, such differences may reflect artifacts rather than genuine biological effects.

Taken together, these factors undermine the specificity and quantitative accuracy of the DCFH_2 -DA assay. While the probe may provide general insights into oxidative changes within cells, its data must be interpreted within the context of its limitations. Failure to do so risks misattribution of mechanistic effects, such as enhanced fluorescence due to peroxidase activity or redox cycling, rather than true ROS production. We therefore caution researchers against using DCFH_2 -DA as a standalone method for intracellular ROS quantification. Instead, it should be complemented with alternative assays, more selective probes, and appropriate controls to validate findings and ensure biological relevance. Critical assessment of probe chemistry, experimental design, and data interpretation is essential for advancing the accurate study of redox biology in drug discovery and beyond.

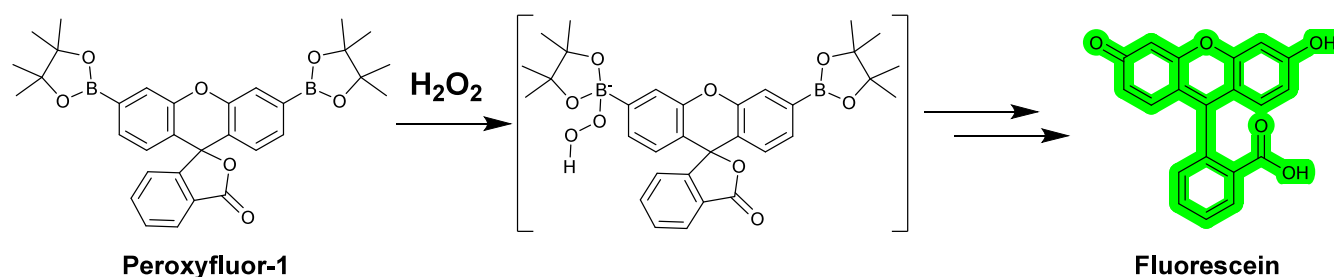


Figure 3. Simplified mechanism of action of arylboronate probes toward the detection of H_2O_2 .

RECOMMENDATION FOR THE DETECTION OF H_2O_2

In simplified experimental systems, H_2O_2 can be detected using an enzyme-catalyzed oxidation reaction with horseradish peroxidase and the probe Amplex Red. In the presence of H_2O_2 , horseradish peroxidase facilitates the oxidation of Amplex Red to resorufin, a fluorescent compound that allows for quantitative measurements (Figure 2).^{54–56} However, this approach is subject to several limitations. Compounds such as ascorbate⁵⁷ and N-acetylcysteine,⁵⁸ which also serve as substrates for horseradish peroxidase, can interfere with the reaction and reduce assay specificity. Moreover, $\text{O}_2^{\bullet-}$ radicals can irreversibly inactivate horseradish peroxidase, thereby compromising the accuracy of the assay. The inclusion of superoxide dismutase, which converts $\text{O}_2^{\bullet-}$ into H_2O_2 and oxygen, can help mitigate this interference by stabilizing horseradish peroxidase activity.⁵⁹ This detection system can be used to measure H_2O_2 efflux from cells, as H_2O_2 can cross cellular membranes either by passive diffusion or by facilitated transport through aquaporin channels. However, extracellular H_2O_2 levels reflect a dynamic interplay between intracellular production, enzymatic degradation by catalase or peroxidases, and membrane permeability. As such, interpretations of extracellular H_2O_2 must account for both cellular and environmental factors influencing its concentration.⁶⁰ Notably, the sensor luminol is also widely used but due to its high reactivity profile, the interpretation of this data can be difficult.⁶¹

For intracellular detection, chemical probes based on arylboronate structures such as Peroxyfluor-1 (MitoPY1, Peroxy Orange-1) have been developed to provide improved selectivity toward H_2O_2 . These compounds rely on a two-electron oxidation mechanism, converting the boronate group into a phenol upon reaction with H_2O_2 (Figure 3). Because this oxidation is specific to two-electron oxidants, these probes are inherently more selective for H_2O_2 than for other ROS that typically act by single-electron transfer.^{62–64} However, the reaction rate between arylboronates and H_2O_2 is relatively slow, which can limit sensitivity and hinder the detection of rapid or localized bursts of H_2O_2 signaling. To address this, newer probe designs incorporating borinic acids have been proposed. These compounds undergo faster oxidation reactions with H_2O_2 , offering improved temporal resolution and the ability to detect lower concentrations. Their enhanced reactivity enables more accurate monitoring of subtle redox changes within specific cellular compartments or under mild oxidative stress conditions.^{65,66} Despite these improvements, boronate- and borinic acid-based probes are not entirely free from cross-reactivity. They can react with other oxidants, such as peroxynitrite and hypochlorous acid, often at rates significantly higher than their reaction with H_2O_2 .⁶⁷ This nonspecific reactivity can confound

the interpretation of results unless appropriate controls are used. Strategies to improve specificity include the use of enzyme inhibitors or scavengers. For instance, catalase and nitric oxide synthase inhibitors can help differentiate between peroxynitrite- and H_2O_2 -related signals by modulating nitric oxide availability.⁶⁸

Genetically encoded fluorescent probes offer a powerful alternative to small-molecule sensors for detecting H_2O_2 . These protein-based probes are expressed directly in cells or organisms, enabling long-term, real-time, and compartment-specific monitoring of H_2O_2 dynamics without the need for external probe administration. HyPer is one of the most widely used genetically encoded H_2O_2 sensors. It consists of a circularly permuted yellow fluorescent protein fused to the OxyR H_2O_2 -sensing domain. Upon exposure to H_2O_2 , OxyR undergoes a conformational change that alters the fluorescence of the protein in a ratiometric manner. HyPer is reversible, allowing detection of transient redox fluctuations. Targeted variants such as HyPer-mito and HyPer-nuc allow for compartment-specific imaging in mitochondria and nuclei, respectively. HyPer has been successfully applied in mammalian cells, zebrafish embryos, and immune cells.^{69,70} roGFP2-Orp1 is another genetically encoded probe that measures H_2O_2 -induced changes in the cellular redox environment. It combines roGFP2, a redox-sensitive GFP, with Orp1, a yeast peroxidase that specifically transfers oxidative equivalents from H_2O_2 to the GFP domain. This causes a shift in the fluorescence excitation ratio, providing a ratiometric and quantitative readout of redox status. roGFP2-Orp1 is particularly useful for monitoring glutathione redox balance, a key aspect of cellular oxidative stress responses.^{71,72}

Electrochemical sensors provide another robust alternative, enabling real-time, quantitative detection of H_2O_2 by measuring the current generated during its electrochemical oxidation or reduction at specially designed electrode surfaces. These electrodes are often functionalized with catalytic materials, such as platinum nanoparticles or enzymes, to enhance sensitivity and selectivity. Electrochemical H_2O_2 sensors have been successfully applied for monitoring oxidative stress in various biological fluids, including blood, saliva, and urine. Notably, implantable electrochemical sensors can track dynamic H_2O_2 fluctuations in tissues such as the brain during pathological conditions like ischemia-reperfusion injury or neurodegenerative diseases.^{73,74}

RECOMMENDATION FOR THE DETECTION OF $\cdot\text{OH}$

Hydroxyphenyl fluorescein (HPF) or its amino derivative aminophenyl fluorescein are widely used fluorescent probe for the detection of $\cdot\text{OH}$ with high sensitivity in live or fixed cells. HPF is a nonfluorescent compound that undergoes a specific chemical transformation upon interaction with $\cdot\text{OH}$. Structurally, it consists of a fluorescein backbone bearing a 2-

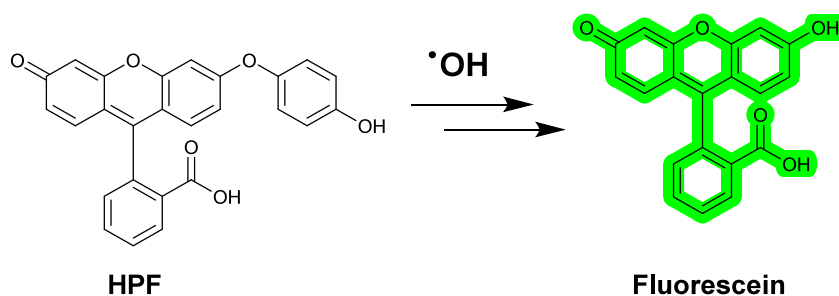


Figure 4. Activation of HPF probe toward the detection of $\bullet\text{OH}$.

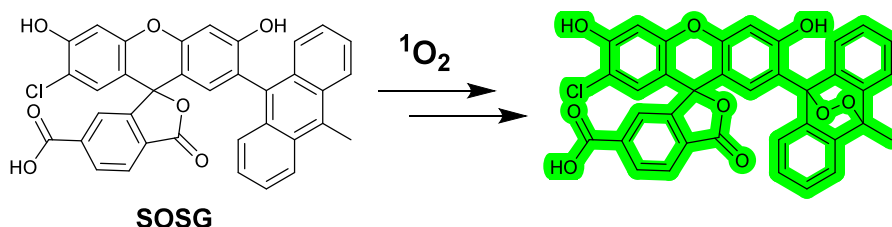


Figure 5. Activation of SOSG probe toward the detection of $^1\text{O}_2$.

hydroxyphenyl substituent. In its native state, HPF exhibits minimal fluorescence due to the quenching effect of the hydroxyphenyl group, which disrupts the π -conjugation necessary for fluorescein excitation and emission. Upon encountering hydroxyl radicals, HPF undergoes oxidation at the hydroxyphenyl moiety. This oxidative reaction results in the cleavage of the 2-hydroxyphenyl substituent that emissive properties of fluorescein (Figure 4).^{75,76} This reaction mechanism is highly selective for hydroxyl radicals. HPF exhibits negligible reactivity toward other common reactive oxygen and nitrogen species, such as H_2O_2 , $\text{O}_2^{\bullet-}$, or ONOO^- . As a result, HPF provides a reliable method for distinguishing hydroxyl radical activity from broader oxidative stress responses in complex biological systems.⁷⁷ Notably, $\bullet\text{OH}$ can also be detected in cell-free systems using small-molecule probes such as terephthalic acid, which reacts with $\bullet\text{OH}$ to form 2-hydroxyterephthalic acid,⁷⁸ or coumarin-3-carboxylic acid, which is converted by $\bullet\text{OH}$ into the fluorescent 7-hydroxycoumarin derivative.^{79,80}

As complementary methods, $\bullet\text{OH}$ can react with salicylate or its analogs to form hydroxylated products such as 2,3- or 2,5-dihydroxybenzoic acid, which can be quantitatively analyzed by high pressure liquid chromatography or liquid chromatography mass spectrometry.^{81,82} Additionally, $\bullet\text{OH}$ reacts with the spin trap 5,5-dimethyl-1-pyrroline N-oxide to form stable adducts, which are detected using electron spin resonance spectroscopy.^{83,84} Another approach involves cyclic hydroxylamine probes, which are oxidized by with $\bullet\text{OH}$, to produce nitroxide radicals that can be measured by electron spin resonance spectroscopy or indirectly by fluorescence using secondary detection reagents.^{85,86}

RECOMMENDATION FOR THE DETECTION OF $^1\text{O}_2$

Singlet Oxygen Sensor Green (SOSG) is a widely used fluorescent probe designed for the selective detection of $^1\text{O}_2$, functioning through a highly specific chemical transformation that results in a fluorescence turn-on signal. Structurally, SOSG is based on a fluorescein core conjugated to an anthracene derivative that serves as the reactive moiety. In its native form, SOSG is either nonfluorescent or exhibits very weak

fluorescence due to intramolecular quenching interactions between the anthracene unit and the fluorescein core. These interactions suppress the emissive properties of the molecule, rendering it effectively dark in the absence of $^1\text{O}_2$. The key mechanism of action of SOSG involves the selective reaction of $^1\text{O}_2$ with the anthracene moiety via a $[4 + 2]$ cycloaddition, leading to the formation of a stable endoperoxide. This chemical conversion disrupts the quenching interactions that had previously inhibited the fluorescein fluorescence, resulting in a marked increase in green fluorescence emission (Figure 5). The reaction is highly selective for $^1\text{O}_2$ and does not proceed with other reactive oxygen species such as H_2O_2 , $\cdot\text{OH}$, or $\text{O}_2\cdot^-$, making SOSG a highly reliable indicator of $^1\text{O}_2$ presence. The fluorescence signal is irreversible and cumulative, meaning it reflects the total amount of $^1\text{O}_2$ encountered over time rather than providing real-time dynamic data. This property makes SOSG particularly useful for end point assays and comparative studies of $^1\text{O}_2$ production.^{87–89} It has found broad applications in photodynamic therapy research, where the generation of $^1\text{O}_2$ by photosensitizers is a key therapeutic mechanism, as well as in studies of photooxidative stress in cells and photoactive materials.^{18,90} Despite its advantages, SOSG has several limitations. One major concern is the potential for photoactivation artifacts under strong illumination, especially in the presence of light and oxygen, SOSG itself can generate $^1\text{O}_2$, leading to self-amplified fluorescence signals and possible misinterpretation of results. Another issue is its irreversibility; once the anthracene moiety reacts with $^1\text{O}_2$, it cannot return to its original state, thereby precluding multiple cycles of detection or kinetic measurements.^{91,92} Therefore, while SOSG remains a powerful and convenient tool for $^1\text{O}_2$ detection in chemical and biological systems, careful experimental design, appropriate controls, and awareness of its limitations are essential to ensure accurate interpretation of data.

Anthracene derivatives, such as 9,10-diphenylanthracene, rely on a well-characterized [4 + 2] cycloaddition reaction between the anthracene moiety and singlet oxygen, forming a stable endoperoxide. This reaction causes a distinct loss of the native fluorescence of the anthracene group, and thus detection is based on fluorescence quenching. The mechanism is highly

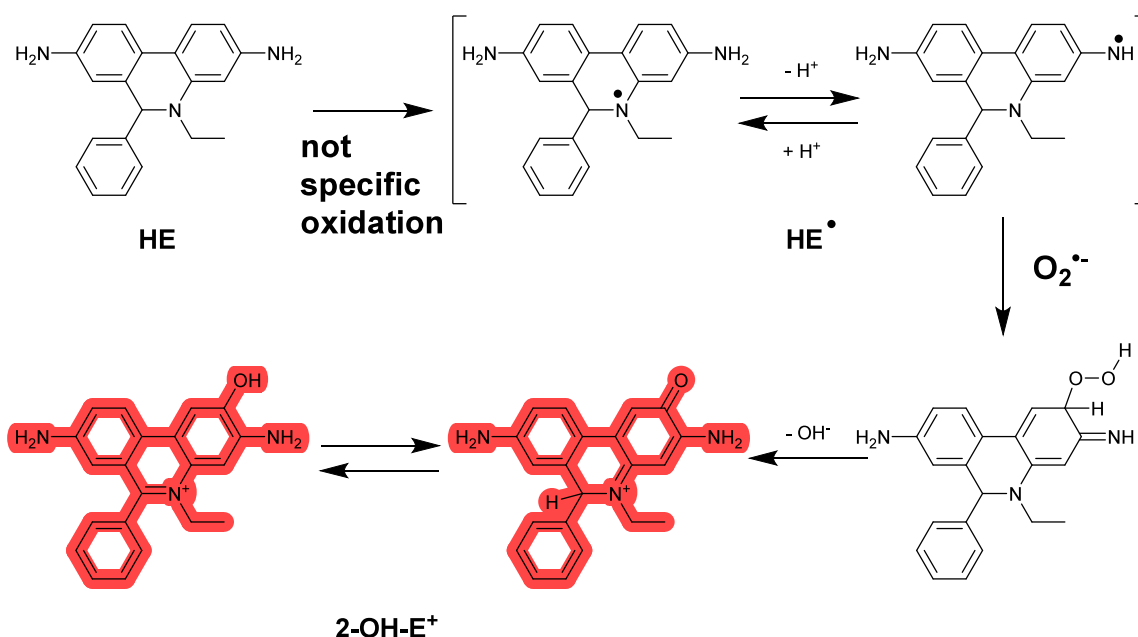


Figure 6. Activation of HE probe toward the detection of O₂^{•-}.

specific to ¹O₂ and has been extensively used in solution-phase assays to quantify singlet oxygen production in photochemical systems. However, despite their chemical reliability, these probes suffer from poor water solubility and lack of inherent cell permeability. Moreover, the quenching-based signal (a decrease in fluorescence) complicates quantitative analysis in biological systems where autofluorescence and photobleaching can also lead to signal reduction. Consequently, anthracene-based probes are not ideally suited for cellular imaging of singlet oxygen.^{93,94}

BODIPY-based probes represent a versatile and tunable class of ¹O₂ sensors. In these probes, a BODIPY fluorophore is chemically modified by incorporating singlet oxygen-reactive functional groups, such as anthracene or electron-rich alkenes, that modulate the fluorescence upon reaction with ¹O₂. Depending on the design, BODIPY-based probes can exhibit fluorescence turn-on, quenching, or ratiometric responses. Their high photostability, sharp emission spectra, and structural flexibility make them attractive for various imaging applications. Furthermore, BODIPY dyes can be engineered for improved water solubility and organelle targeting, enhancing their potential for cellular studies. Despite these advantages, BODIPY-based probes may still face limitations including synthetic complexity, susceptibility to photobleaching under intense illumination, generation of ¹O₂ by the probe itself, and the potential for off-target reactions with other oxidants. Moreover, the fluorescence response is often sensitive to the surrounding microenvironment, which may complicate interpretation of results in heterogeneous biological systems.^{95,96}

In addition to fluorescence-based probes, several complementary methods are available for the detection and quantification of ¹O₂. These techniques are often used alongside fluorescence methods to validate results, provide quantitative data, or gain mechanistic insights. One of the most specific approaches is time-resolved near-infrared phosphorescence detection. ¹O₂ emits weak phosphorescence at ~ 1270 nm as it relaxes to the ground state, which can be captured using highly sensitive detectors. This method offers direct, real-time monitoring with excellent specificity. However, the weak

emission and need for specialized, expensive equipment limit its use, especially in biological samples where signal loss due to scattering and absorption is significant.^{97,98} Electron spin resonance spectroscopy with spin-trapping is another highly selective method. ¹O₂ reacts with spin traps like (2,2,6,6-Tetramethylpiperidin-1-yl)oxyl to form stable paramagnetic species detectable by electron spin resonance spectroscopy. This allows quantitative detection and discrimination from other ROS. However, the approach is largely limited to in vitro systems due to limited cell permeability of spin traps and the complexity of electron spin resonance spectroscopy instrumentation.^{99,100} Chemical trapping methods use reactive substrates such as 1,3-diphenylisobenzofuran or furfuryl alcohol, which react with ¹O₂ to form stable products. These changes can be monitored by UV-vis spectroscopy or quantified by high pressure liquid chromatography or liquid chromatography mass spectrometry. These techniques are simple and cost-effective but offer only end point data and limited selectivity unless paired with selective traps.^{101–103}

RECOMMENDATION FOR THE DETECTION OF O₂^{•-}

Intracellular O₂^{•-} is most often detected using dihydroethidium, also referred to as hydroethidine (HE). A mitochondria-targeted analogue of this probe, known as MitoSOX, is also widely employed for superoxide measurements in the mitochondria. These reduced probes are essentially nonfluorescent until they undergo oxidation, which makes them attractive for live-cell studies of oxidative stress. The oxidation of HE is initiated by a one-electron transfer, a step that may be catalyzed by a variety of ROS, enzymatic redox systems, or even photooxidative processes. Subsequently, only the reaction with superoxide O₂^{•-} triggers a highly specific secondary step involving hydroxylation at the 2-position of the ethidium chromophore. This targeted chemical transformation produces 2-hydroxyethidium (2-OH-E⁺). The hydroxylation event not only alters the molecular structure but also significantly changes the photo-physical properties of the probe, producing a characteristic fluorescence signal (Figure 6).¹⁰⁴ Upon intercalation into DNA, the fluorescence intensity of 2-OH-E⁺ increases markedly,

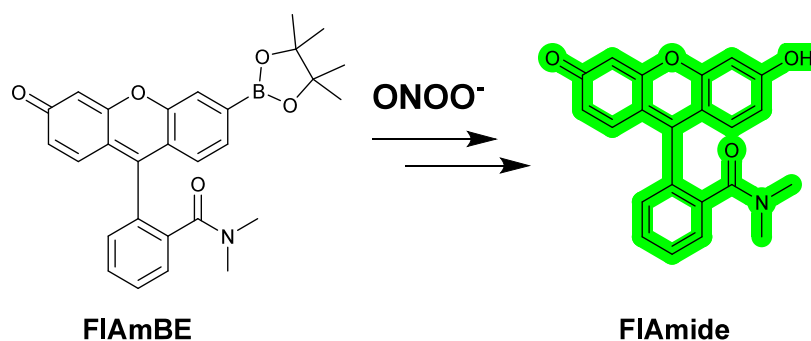


Figure 7. Activation of an arylboronate probe toward the detection of ONOO^- .

further enhancing signal detection. Despite this specificity at the chemical level, fluorescence-based measurements using HE or MitoSOX are prone to significant artifacts. Both probes can also be oxidized nonspecifically by other reactive oxygen species, enzymatic redox systems, or photooxidative processes to form ethidium (E^+), a product that is not diagnostic of superoxide. Importantly, the fluorescence spectra of E^+ and 2-OH-E^+ overlap extensively, which makes it difficult to separate their contributions to the overall signal when using standard fluorescence microscopy or bulk fluorescence spectroscopy. As a result, measuring fluorescence alone often overestimates superoxide levels, since the detected signal may be dominated by nonspecific oxidation products rather than the superoxide-specific 2-OH-E^+ .^{105,106} To overcome this limitation, accurate quantification of $\text{O}_2^{\bullet-}$ requires analytical separation of the oxidation products. Liquid chromatography–mass spectrometry provides the most reliable means of distinguishing E^+ and 2-OH-E^+ and other oxidation products, by resolving them based on retention time and mass-to-charge ratio. Without such separation, it is difficult to assess the true contribution of superoxide-dependent oxidation to the observed fluorescence.^{107,108} Several additional factors can complicate interpretation of results obtained with HE or MitoSOX. The cellular uptake of these probes, and in the case of MitoSOX its mitochondrial targeting efficiency, determine how much probe is available for reaction with superoxide. Uneven intracellular distribution can bias detection toward regions of higher probe concentration, while the formation of multiple oxidation products beyond E^+ and 2-OH-E^+ can further confound measurements.¹⁰⁹ Moreover, both E^+ and 2-OH-E^+ bind strongly to DNA, which substantially increases their fluorescence independently of superoxide concentration, potentially overestimating the respective $\text{O}_2^{\bullet-}$ levels. In response to these issues, modified probes such as NeoD and MitoNeoD have been developed. These analogues contain structural modifications to the HE core that prevent DNA intercalation, thereby reducing fluorescence enhancement artifacts while retaining reactivity toward superoxide.¹¹⁰ Such advances improve the reliability of superoxide detection in complex biological systems, but even with these improved probes, product identification and quantification by LC–MS remains the gold standard.

Lucigenin is a chemiluminescent probe that emits light upon reaction with superoxide. The mechanism involves superoxide reducing lucigenin to form an unstable radical cation intermediate, which subsequently decomposes to release photons. This chemiluminescence signal is proportional to the amount of superoxide present, making lucigenin highly sensitive for detecting $\text{O}_2^{\bullet-}$ in solution. Lucigenin has been widely used to measure extracellular $\text{O}_2^{\bullet-}$ production. However, a major

limitation is that lucigenin can undergo redox cycling in the presence of cellular reductants or enzymes, which may artificially generate $\text{O}_2^{\bullet-}$ and thereby lead to overestimation of ROS levels. Additionally, lucigenin primarily detects superoxide accessible in extracellular spaces, rather than directly measuring intracellular $\text{O}_2^{\bullet-}$. The chemiluminescent signal is also susceptible to quenching or interference by other biological molecules.^{111–113}

Hydrocyanines, which are structurally related to hydroethidine but modified to enhance photostability and fluorescence intensity, represent another important class of superoxide probes. Like HE, hydrocyanines undergo oxidation via a one-electron transfer catalyzed by reactive oxygen species or redox enzymes, producing fluorescent cyanine products. Superoxide may specifically induce hydroxylation or oxidation steps leading to distinct fluorescent species, although this specificity is generally less well characterized than for HE. Hydrocyanines often exhibit brighter fluorescence and greater photostability, making them attractive for imaging applications. Nonetheless, these probes also suffer from nonspecific oxidation by other reactive species and enzymatic systems. The fluorescence spectra of specific and nonspecific oxidation products often overlap, complicating data interpretation. Furthermore, oxidation products of hydrocyanines can intercalate into DNA, which dramatically enhances fluorescence intensity and introduces artifacts. Therefore, accurate quantification of superoxide with these probes typically requires chromatographic separation techniques such as high-performance liquid chromatography (HPLC) or liquid chromatography–mass spectrometry (LC–MS) to distinguish specific products.^{114–118}

The cytochrome c reduction assay offers a classical biochemical approach to superoxide detection. In this assay, superoxide reduces ferricytochrome c to ferrocytochrome c, leading to changes in absorbance that can be measured spectrophotometrically. Some adaptations incorporate fluorescent derivatives or coupled systems to increase sensitivity. This method provides a direct measure of superoxide production and is well-established in biochemical research. It is frequently used with cell suspensions, isolated mitochondria, or enzymatic systems. However, the assay does not inherently distinguish intracellular from extracellular superoxide and can be affected by other cellular reductants or enzymes capable of reducing cytochrome c independently of superoxide. Additionally, it lacks spatial and temporal resolution, making it poorly suitable for live-cell imaging or localized measurements of reactive oxygen species.^{119–121}

RECOMMENDATION FOR THE DETECTION OF ONOO[−]

ONOO[−] is a highly reactive nitrogen species with complex chemistry that contributes to oxidative and nitrative stress in biological systems. One of its major physiological reactions occurs with carbon dioxide, a ubiquitous molecule in living systems due to the CO₂/HCO₃[−] buffering system.¹²² This reaction rapidly yields two potent secondary species: the carbonate radical anion CO₃^{•−} and the nitrogen dioxide radical •NO₂. Both are capable of modifying a broad range of biomolecules and are known to interfere with commonly used ROS probes. Among available detection strategies, boronate-based probes stand out due to their exceptional reactivity with ONOO[−], nearly a million times faster than their reaction with H₂O₂. This high kinetic selectivity allows boronate probes to serve as effective tools for assessing ONOO[−] formation under appropriate experimental conditions (Figure 7).¹²³

CONCLUSIONS

In conclusion, the reliable detection and quantification of reactive oxygen species remain a critical yet challenging aspect of understanding their multifaceted roles in biology and disease. Although significant progress has been made in the development of chemical and biological probes, the inherent instability, high reactivity, and diversity of ROS continue to limit precise measurements. While herein the application of fluorescent probes is discussed, the use of multiple analytical techniques is recommended to achieve a more comprehensive understanding of the specific types of ROS produced and to verify their presence accurately. By critically evaluating current probe technologies and outlining practical considerations for probe selection, this perspective article provides a strategic framework to enhance experimental accuracy and reproducibility. To mitigate the limitations associated with fluorescent probes, ROS measurements should be supplemented with orthogonal methodologies. Complementary techniques such as electron paramagnetic resonance spectroscopy with spin trapping, mass spectrometry-based approaches, electrochemical sensors, genetically encoded redox reporters, and chemiluminescence assays have been shown to provide valuable. The combined application of these methods can improve confidence in ROS identification, enable cross-validation of results, and reduce probe-related artifacts. Ultimately, improving ROS detection methodologies will not only deepen our understanding of their regulatory functions in cellular processes but also accelerate the development of targeted therapeutic strategies for ROS-associated diseases.

SIGNIFICANCE

Accurate measurement of ROS is essential for understanding their roles in physiology and disease. This article highlights the limitations of the commonly used, nonselective probe DCFH-DA and provides practical guidance for selecting and applying fluorescent probes. By enabling more reliable detection of specific ROS in complex biological systems, these recommendations enhance the rigor and reproducibility of redox chemistry in biological contexts and medical research.

AUTHOR INFORMATION

Corresponding Author

Johannes Karges — Faculty of Chemistry and Biochemistry,
Ruhr-University Bochum, 44780 Bochum, Germany;

orcid.org/0000-0001-5258-0260; Phone: +49
2343224187; Email: johannes.karges@ruhr-uni-bochum.de

Complete contact information is available at:

<https://pubs.acs.org/10.1021/acs.jmedchem.5c03494>

Notes

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ABBREVIATIONS

CO₃^{•−}, carbonate radical DCF, 2',7'-dichlorofluorescein; DCFH2-DA, 2',7'-dichlorofluorescein diacetate; DNA, DNA; H₂O₂, hydrogen peroxide; HE, hydroethidine; HPF, hydroxyphenyl fluorescein; •NO₂, nitrogen dioxide radical; •OH, hydroxyl radical; ¹O₂, singlet oxygen; O₂^{•−}, superoxide anion; ONOO[−]/ONOOH, peroxynitrite; ROS, reactive oxygen species; SOSG, singlet oxygen sensor green

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